

Industrial Internship Report on "HVAC and Remote Energy Monitoring System based on DAMS"

Prepared by

MAJETI JASWANTH SAI

Executive Summary

This report presents the work carried out during the six-week Industrial Internship organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship focused on understanding the Digital Acquisition and Monitoring System (DAMS) and developing a simulated HVAC and Remote Energy Monitoring System using Node-RED without the use of physical hardware.

The project involved designing a complete IoT-based monitoring workflow consisting of HVAC data simulation, MQTT-based communication, real-time dashboard visualization, SQLite database integration, historical data storage, and analytics. A FlowFuse Dashboard was developed to display live environmental and electrical parameters, while historical records and key performance indicators were generated using SQL-based analytics.

Throughout the internship, practical exposure was gained in Node-RED, MQTT, Mosquitto, SQLite, dashboard development, and industrial IoT architecture. The project demonstrates how an industrial monitoring solution can be simulated and implemented using software tools while following a modular IoT architecture. The internship provided valuable hands-on experience in solving an industry-oriented problem and strengthened both technical and problem-solving skills.

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1 Preface

Industrial internships play an important role in bridging the gap between academic learning and industry practices by providing students with practical exposure to real-world engineering problems. This six-week internship, organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT), provided an opportunity to understand the concepts of Industrial Internet of Things (IIoT) and their application in monitoring systems.

During the internship, I worked on the project titled **"HVAC and Remote Energy Monitoring System based on DAMS"**, where the objective was to simulate an industrial monitoring solution using Node-RED without physical hardware. The project involved designing a complete IoT workflow that generates HVAC sensor data, transmits it through the MQTT protocol, stores it in an SQLite database, and visualizes both real-time and historical information on an interactive dashboard. In addition, analytical insights were generated from the stored data to demonstrate business-oriented monitoring capabilities.

The internship was planned over six weeks, beginning with understanding the DAMS platform and monitoring requirements, followed by learning Node-RED and MQTT communication. The later stages focused on system design, dashboard development, database integration, data analytics, testing, and documentation.

This internship significantly enhanced my understanding of Industrial IoT architecture, Node-RED application development, MQTT communication, database integration, and dashboard visualization. It also improved my technical problem-solving skills and provided valuable insight into how software-based monitoring solutions are designed and implemented in industrial environments.

I sincerely express my gratitude to UniConverge Technologies Pvt. Ltd., upskill Campus, The IoT Academy, and all the mentors who guided and supported me throughout this internship. Their guidance and encouragement made this learning experience both valuable and rewarding.

I hope this report accurately presents the work carried out during the internship and serves as a reflection of the knowledge and practical experience gained throughout the program.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoraWAN), Java Full Stack, Python, Front end** etc.



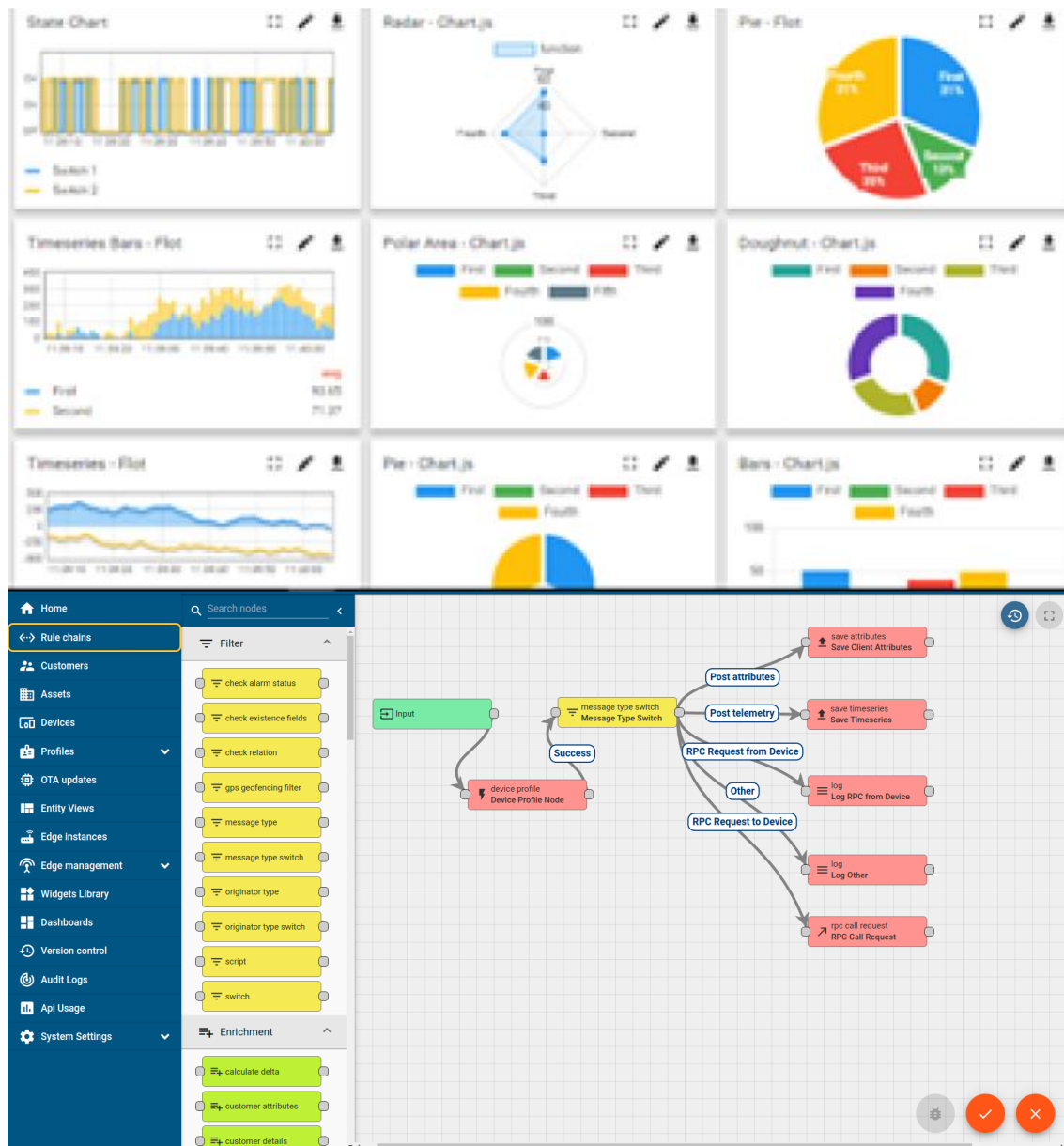
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

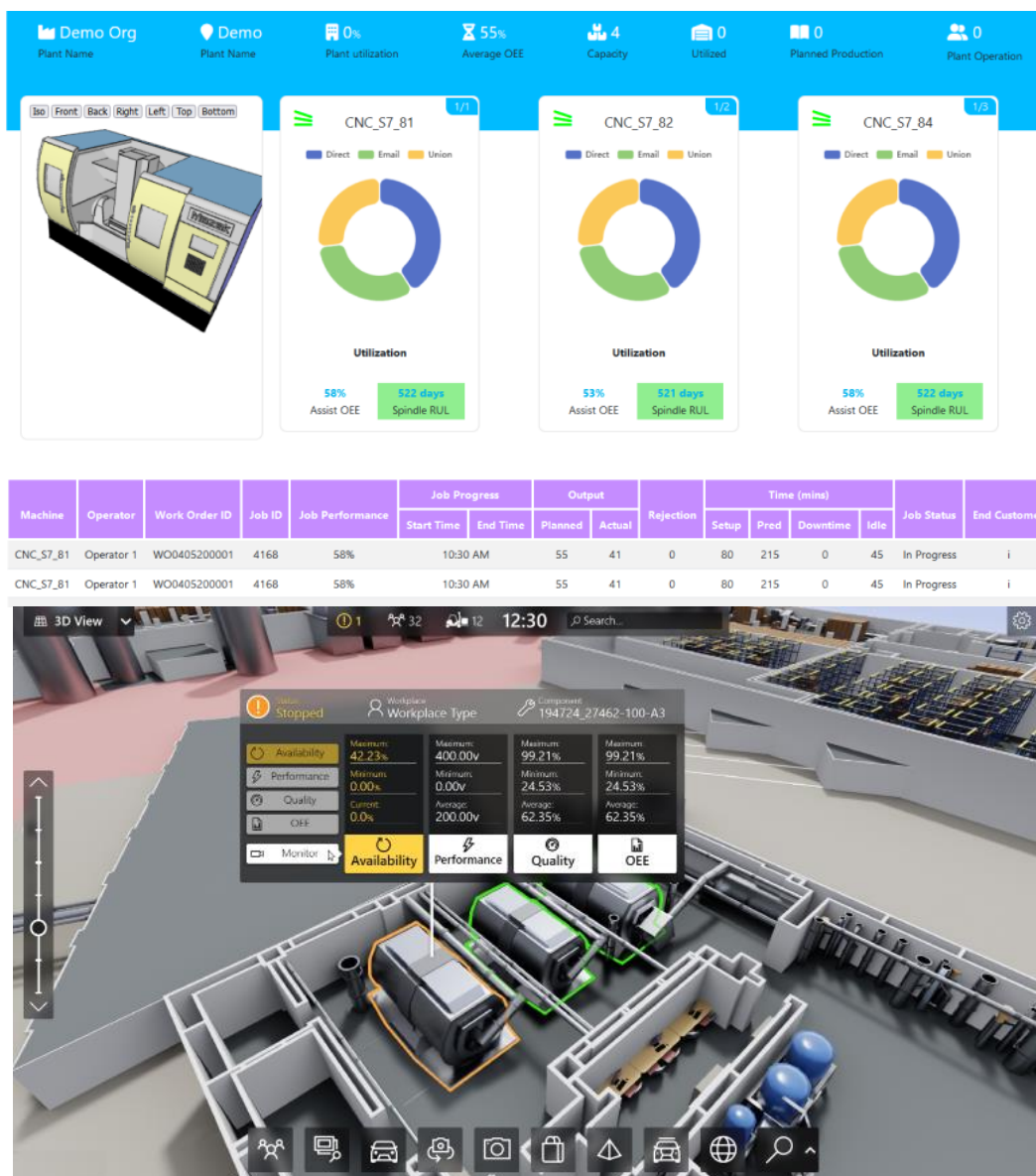
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



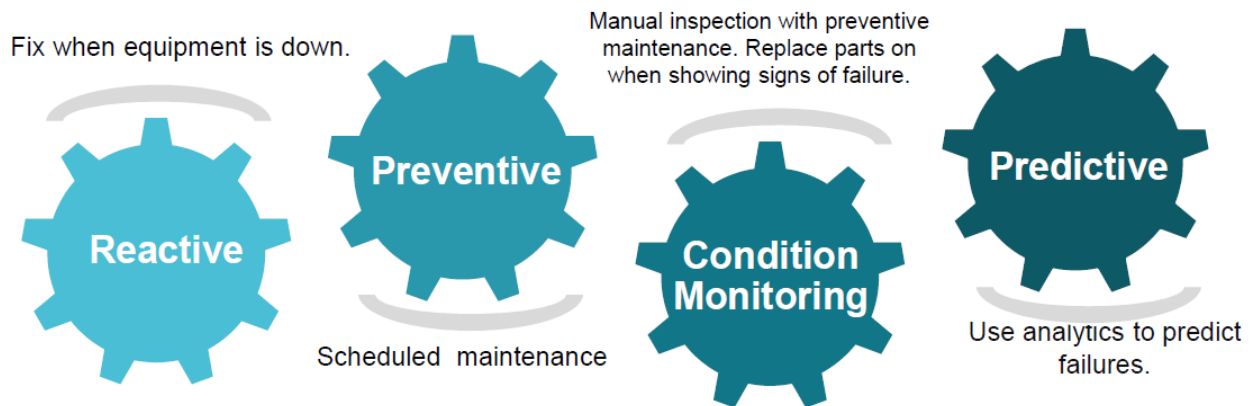


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

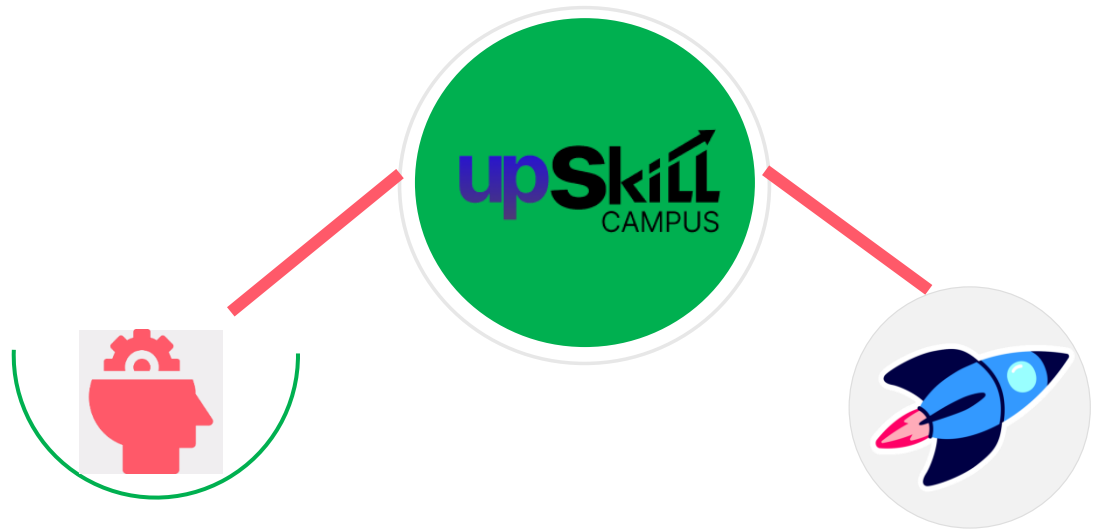
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

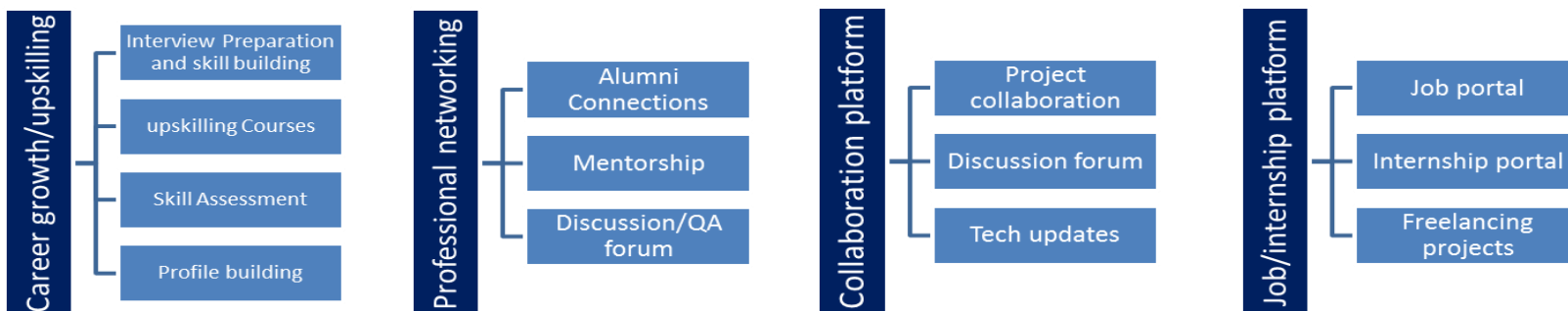
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Reference

- [1] Node-RED. *Node-RED Documentation*. Available: <https://nodered.org/docs/>
- [2] Eclipse Foundation. *MQTT Essentials*. Available: <https://mqtt.org/>
- [3] Eclipse Mosquitto. *Mosquitto Documentation*. Available: <https://mosquitto.org/documentation/>

2.5 Glossary

Terms	Description
HVAC	Heating, Ventilation, and Air Conditioning
DAMS	Digital Acquisition and Monitoring System
MQTT	Message Queuing Telemetry Transport – a lightweight messaging protocol for IoT communication
Node-RED	A flow-based programming tool used for wiring together hardware devices, APIs, and online services
Mosquitto	An open-source MQTT message broker used for message exchange
Flow	A sequence of interconnected nodes in Node-RED representing an application or process
SQLite	A lightweight, serverless relational database management system

3 Problem Statement

Industrial facilities such as commercial buildings, factories, hospitals, and office complexes rely on Heating, Ventilation, and Air Conditioning (HVAC) systems to maintain comfortable environmental conditions while ensuring efficient energy utilization. Continuous monitoring of HVAC operating parameters is essential for maintaining system performance, identifying abnormal operating conditions, and reducing unnecessary energy consumption.

Traditional monitoring approaches often require dedicated hardware, multiple sensing devices, and specialized monitoring equipment, which may not always be available during the initial stages of system design, training, or prototyping. Therefore, there is a need for a software-based monitoring solution that can simulate an Industrial Internet of Things (IIoT) environment and demonstrate the complete data acquisition and monitoring workflow without relying on physical hardware.

The problem statement selected during this internship was to design and develop a **HVAC and Remote Energy Monitoring System based on the Digital Acquisition and Monitoring System (DAMS)** using Node-RED. The objective was to simulate HVAC sensor data, transmit the generated telemetry using the MQTT communication protocol, visualize real-time information on a dashboard, store historical records in an SQLite database, and generate analytical insights from the collected data.

The monitoring system was designed to continuously observe the following parameters:

- Temperature (°C)
- Humidity (%)
- Voltage (V)
- Current (A)
- Energy Consumption
- Occupancy Count

The complete solution was implemented as a software simulation without the use of physical sensors or embedded hardware, following the internship guidelines for a DAMS-based Industrial IoT monitoring system.

4 Existing and Proposed solution

Traditional HVAC and energy monitoring systems are widely used in industries to supervise environmental and electrical parameters. These systems typically rely on physical sensors, programmable logic controllers (PLCs), data acquisition hardware, and Supervisory Control and Data Acquisition (SCADA) systems to collect and process real-time information. The acquired data is transmitted to centralized monitoring platforms for visualization, analysis, and maintenance.

Although these systems provide reliable monitoring, they have several limitations during development and training:

- Require dedicated sensing and data acquisition hardware.
- Higher deployment and maintenance costs.
- Difficult to prototype or test without physical equipment.
- Limited accessibility for students and beginners learning Industrial IoT concepts.
- Hardware failures or communication issues can interrupt testing and experimentation.

To overcome these limitations, a software-based Industrial IoT monitoring system was developed using **Node-RED** to simulate the complete HVAC monitoring workflow without requiring physical hardware.

The proposed solution consists of the following modules:

- HVAC sensor data simulation using Node-RED Function nodes.
- Data validation and processing before transmission.
- MQTT-based communication using the Mosquitto broker.
- Real-time dashboard for monitoring environmental and electrical parameters.
- SQLite database for storing historical sensor data.
- SQL-based analytics for generating key performance indicators.
- Rule-based system status monitoring for identifying normal, warning, and critical operating conditions.

The system continuously simulates and monitors:

- Temperature
- Humidity
- Voltage
- Current
- Energy
- Occupancy

The developed solution provides several advantages:

- Eliminates the need for physical hardware during development and learning.
- Demonstrates a complete Industrial IoT monitoring architecture.
- Enables real-time visualization of HVAC parameters.
- Stores historical data for future analysis.
- Generates useful analytical insights using SQL queries.
- Provides a modular architecture that can be extended to integrate real sensors, cloud platforms, and predictive analytics in future deployments.

4.1 Project Repository (Github link)

<https://github.com/jaswanthmajeti/upskillcampus>

4.2 Report submission (Github link) :

https://github.com/jaswanthmajeti/upskillcampus/blob/main/HVACAndRemoteEnergyMonitoringSystem_JASWANTH_USC_UCT.pdf

5 Proposed Design/ Model

The proposed HVAC and Remote Energy Monitoring System was developed as a software-based Industrial Internet of Things (IIoT) solution using Node-RED. The system simulates HVAC sensor data, processes the generated telemetry, transmits it through the MQTT protocol, stores historical records in an SQLite database, and presents both real-time and historical information through an interactive dashboard. The modular design makes the system easy to understand, maintain, and extend for real industrial deployments.

5.1 High Level Diagram

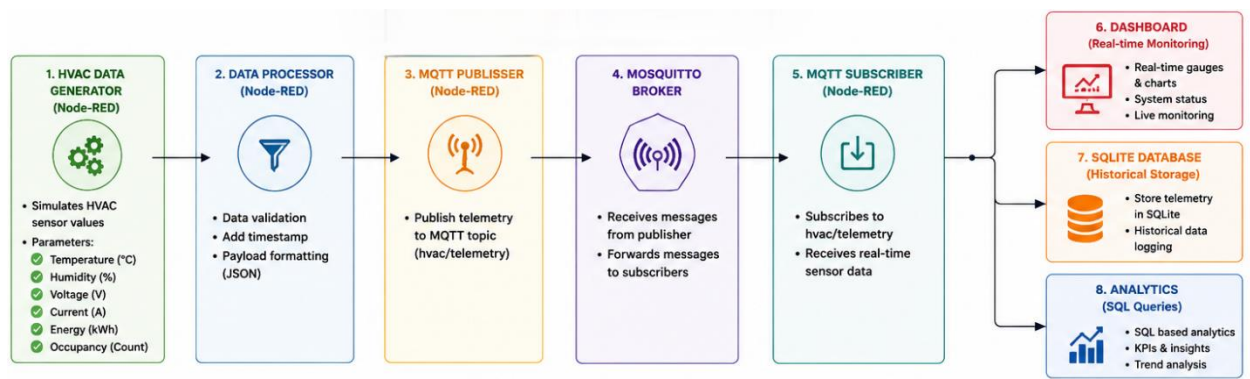


Figure 5.1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

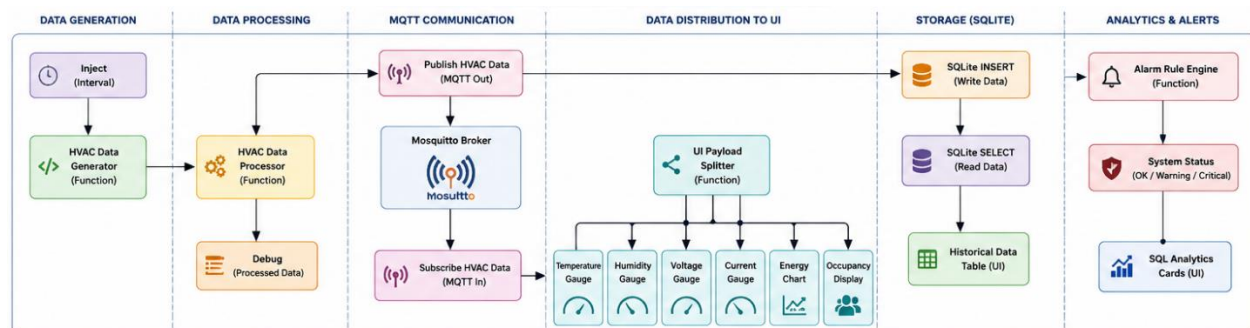


Figure 5.2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The proposed monitoring system consists of the following interfaces.

- **1. Data Generation Interface**
 - Simulates HVAC sensor readings.
 - Generates Temperature, Humidity, Voltage, Current, Energy, and Occupancy values.
 - Produces realistic telemetry without physical sensors.

- **2. Data Processing Interface**
 - Validates incoming sensor values.
 - Adds timestamps.
 - Standardizes the telemetry payload before transmission.

- **3. MQTT Communication Interface**
 - Uses the MQTT publish–subscribe communication model.
 - Mosquitto acts as the message broker.
 - Enables reliable data transfer between system components.

- **4. Database Interface**
 - Stores processed telemetry in an SQLite database.
 - Supports historical data retrieval.
 - Enables SQL-based analytics.

- **5. Dashboard Interface**

The dashboard provides real-time monitoring through:

- Temperature Gauge
- Humidity Gauge
- Voltage Gauge
- Current Gauge
- Energy Trend Chart
- Occupancy Display
- Historical Data Table
- Analytics Cards
- System Status Alerts

- **6. Analytics Interface**

The analytics module retrieves stored information from the database and computes key performance indicators, including:

- Total Records
- Average Temperature
- Average Humidity
- Total Energy
- Maximum Voltage
- Minimum Voltage

These insights help evaluate HVAC performance and energy usage over time.

6 Performance Test

The developed HVAC and Remote Energy Monitoring System was tested to verify the correctness of data generation, MQTT communication, database storage, dashboard visualization, and analytical functions. Since the project was implemented as a software-based Industrial IoT simulation, the evaluation focused on validating the functional behavior of each system component under normal operating conditions.

6.1 Test Plan/ Test Cases

Test ID	Test Description	Expected Result	Status
TC-01	Generate simulated HVAC data	Valid sensor values are generated	Pass
TC-02	Validate processed telemetry	Timestamp added and payload formatted correctly	Pass
TC-03	Publish data through MQTT	Data successfully published to broker	Pass
TC-04	Subscribe to MQTT topic	Subscriber receives published data	Pass
TC-05	Display real-time dashboard values	Dashboard updates continuously	Pass
TC-06	Store telemetry in SQLite	Records inserted successfully	Pass
TC-07	Retrieve historical records	Recent records displayed correctly	Pass
TC-08	Execute SQL analytics	Analytics cards display correct values	Pass
TC-09	Evaluate alarm conditions	System status changes according to thresholds	Pass

6.2 Test Procedure

The following procedure was followed to validate the system:

1. The Node-RED flow was deployed and executed.
2. The HVAC Data Generator produced simulated sensor values.

3. The Data Processor validated the readings and added timestamps.
4. Processed telemetry was published to the Mosquitto MQTT broker.
5. The subscriber node received the transmitted messages.
6. The dashboard updated the monitored parameters in real time.
7. Sensor records were stored in the SQLite database.
8. SQL queries retrieved historical records and generated analytical metrics.
9. The Alarm Rule Engine evaluated temperature and current thresholds to display the appropriate system status.

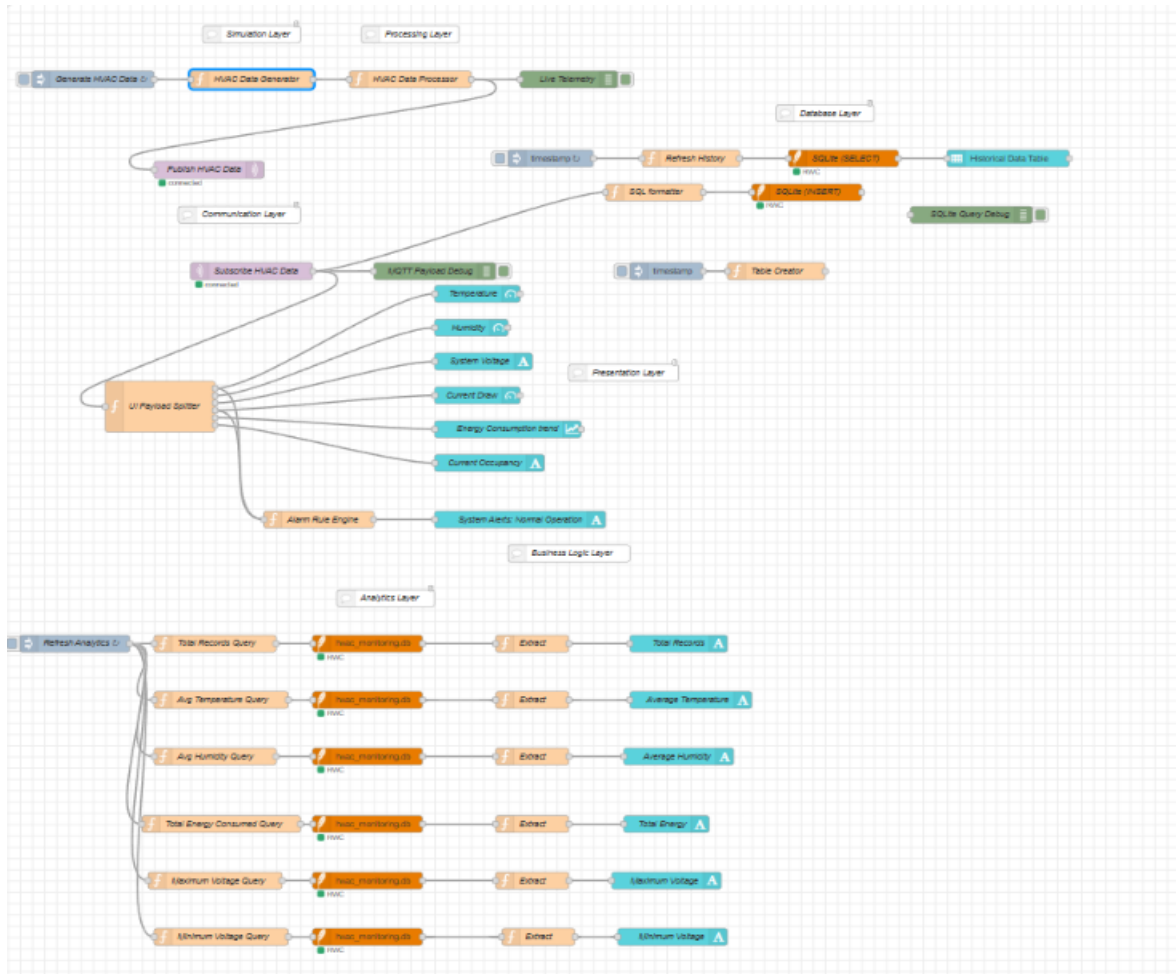
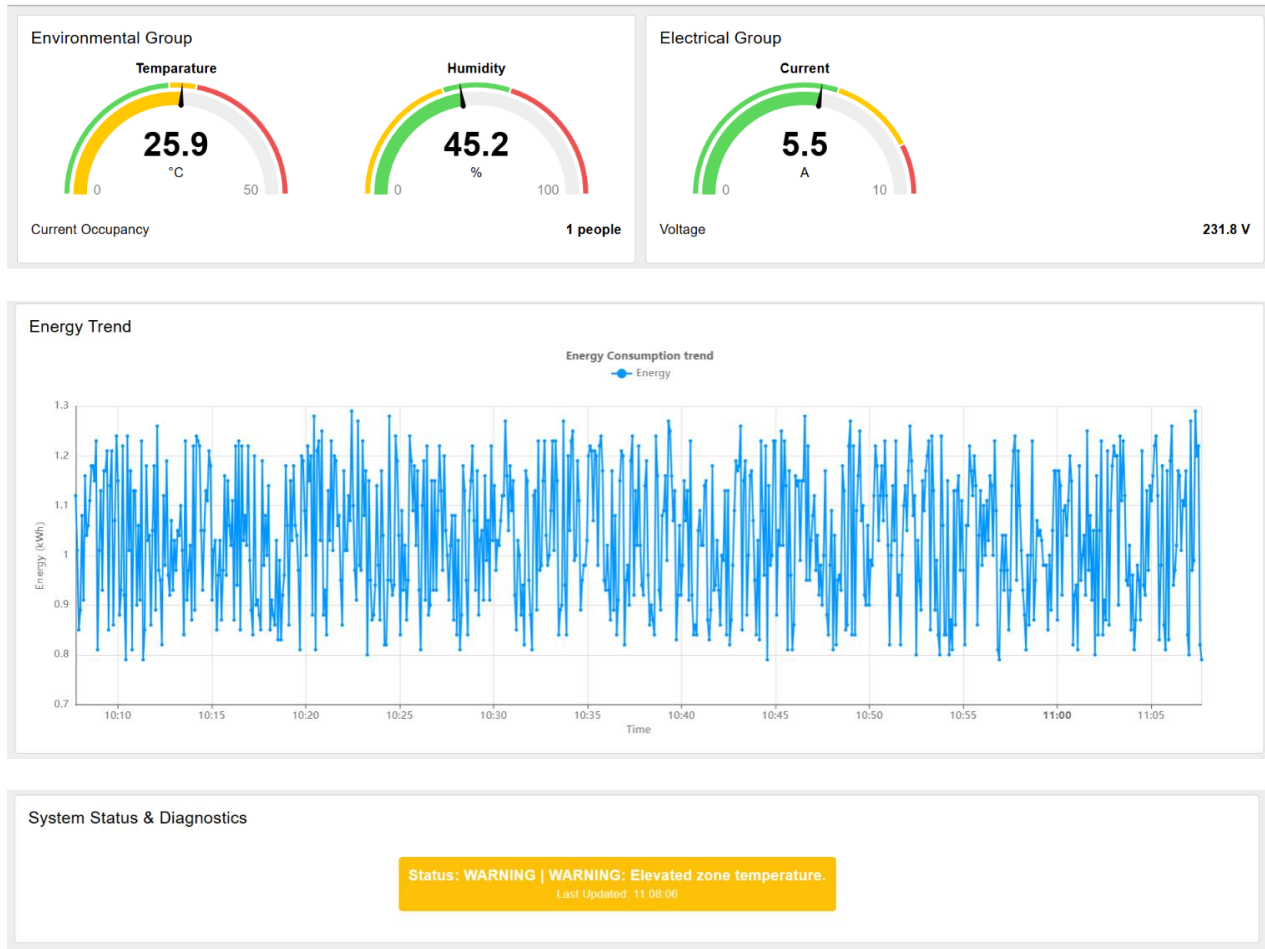


Figure 6.1: Node-RED Flow Executing Successfully

≡ HVAC & Remote Energy Monitoring System



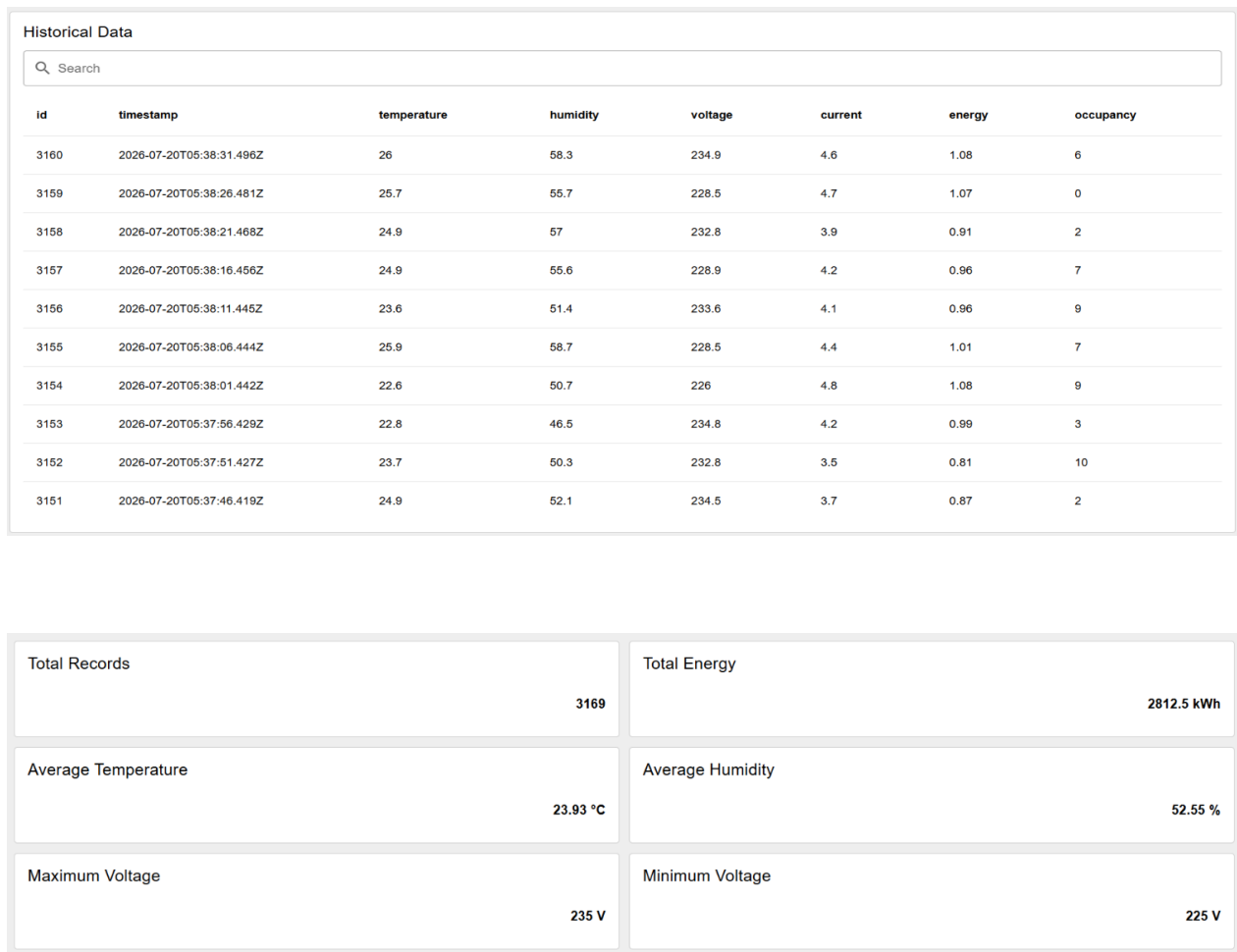


Figure 6.2: Real-time dashboard displaying simulated HVAC parameters.

6.3 Performance Outcome

The developed system successfully performed all intended functions during testing.

The observed outcomes include:

- Continuous generation of simulated HVAC sensor data.
- Successful MQTT communication between publisher and subscriber.
- Reliable real-time dashboard visualization.
- Correct storage of telemetry records in the SQLite database.
- Successful retrieval of historical records.

- Accurate SQL-based analytical calculations.
- Proper evaluation of system status through the alarm rule engine.
- Stable execution of the complete monitoring workflow without functional errors.

Overall, the proposed solution demonstrated the successful implementation of an Industrial IoT monitoring system using Node-RED, MQTT, SQLite, and dashboard visualization in a software simulation environment.

7 My learnings

During this internship, I gained practical knowledge of Industrial Internet of Things (IIoT) concepts by developing a software-based HVAC and Remote Energy Monitoring System. The project provided hands-on experience in designing, implementing, testing, and documenting an Industrial IoT application.

The key learnings from this internship include:

- Understanding the fundamentals of the Digital Acquisition and Monitoring System (DAMS).
- Learning flow-based application development using Node-RED.
- Implementing MQTT-based communication using the Mosquitto broker.
- Simulating HVAC sensor data without physical hardware.
- Designing real-time monitoring dashboards for industrial applications.
- Working with SQLite for data storage and retrieval.
- Writing SQL queries to generate analytical insights from stored telemetry.
- Developing a modular system architecture for Industrial IoT applications.
- Understanding the complete workflow of data generation, communication, storage, visualization, and analysis.
- Improving technical documentation, problem-solving, and project implementation skills.

Overall, this internship enhanced both my technical knowledge and practical understanding of Industrial IoT system development and prepared me for future work in embedded systems, IoT, and industrial automation.

8 Future Work Scope

The developed HVAC and Remote Energy Monitoring System provides a foundation for Industrial IoT monitoring. The following enhancements can be considered in future work:

- Integrate real hardware sensors instead of simulated data.
- Connect microcontrollers such as ESP32 or Arduino for live data acquisition.
- Store telemetry in cloud databases for remote access and scalability.
- Implement predictive maintenance using machine learning techniques.
- Add user authentication and role-based access control for secure monitoring.
- Configure SMS, email, or mobile notifications for abnormal operating conditions.
- Develop a responsive web or mobile application for remote monitoring.
- Support multiple HVAC units within a single monitoring platform.
- Integrate industrial communication protocols such as Modbus or OPC UA.
- Deploy the solution on cloud platforms for real-time remote monitoring.

These enhancements would improve the scalability, reliability, and industrial applicability of the proposed monitoring system.